

Git & GitHub

Joshua Zingale

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The World of Software Development

- ▶ **Building is Iterative:** We don't just write code once; we refine, refactor, and expand it over time.
- ▶ **Complexity:** Even “simple” apps can soon have hundreds of files and thousands of lines of logic.
- ▶ **The Goal:** To deliver stable, functional software while maintaining the ability to change it quickly.

The “Save As” Nightmare

- ▶ **Version Fatigue:** We've all been there: `script_v1.py`, `script_v2_final.py`, `script_v2_final_FIXED.py`.
- ▶ **Lack of Context:** What exactly is different between this and that version?
- ▶ **Risk:** Deleting a block of code to “try something else” often means losing the original work forever if you don't have a backup.

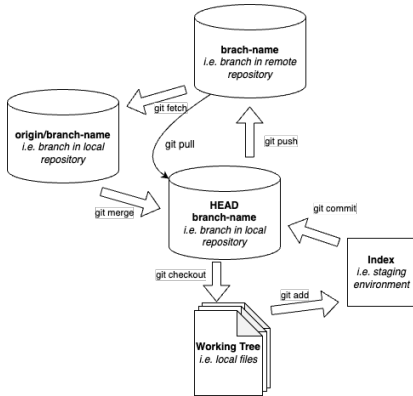
Compounding Issues in a Team

- ▶ **The “Overwriting” Problem:** Two people edit the same file at the same time. Who wins?
- ▶ **The “Broken Master” Problem:** One person’s experimental code breaks the entire project for everyone else.
- ▶ **Communication Overhead:** Constantly asking “Are you working on the login page?” or “Which version is the latest?” slows down development.

Enter: Version Control Systems (VCS)

- ▶ **The Time Machine:** VCS records every change made to the codebase.
- ▶ **The Auditor:** It tracks *who* made the change, *when*, and *why*.
- ▶ **The Safety Net:** It allows you to experiment boldly, knowing you can always revert to a “known good” state.
- ▶ **Git** is the industry standard for distributed version control.

Git Architecture



Typical workflow

Download repo
`git clone <uri>`

Edit file
`vim README.md`

Track file
`git add README.md`

Store fix locally
`git commit -m "grammar"`

Store commit on remote
`git push`

Core Concept: Commits & Rollbacks

- ▶ **The Snapshot:** A “Commit” isn’t just a save point; it’s a snapshot of your entire project at a specific moment.
- ▶ **The Hash:** Each commit has a unique ID.
- ▶ **Rollbacks:** If a new feature introduces a critical bug, you can “checkout” or “revert” to a previous commit instantly.
- ▶ *Safety first: No work is ever truly lost¹.*

¹unless you like to live on the edge and use `git push -f`: don don don...

Core Concept: Branching

- ▶ **Parallel Universes:** Branching allows you to diverge from the main project line (often called `main` or `master`).
- ▶ **Isolated Development:** You can build a “Dark Mode” feature on one branch while a teammate fixes a “Login Bug” on another.
- ▶ **Zero Interference:** Changes in a branch do not affect the main codebase until you are ready.

Core Concept: Merging

- ▶ **Bringing it Together:** Once a feature is finished and tested in its branch, you “Merge” it back into the main line.
- ▶ **Automatic Integration:** Git is smart—it combines changes from different files automatically.
- ▶ **Conflict Resolution:** If two people edited the exact same line, Git pauses and asks you to choose which version to keep.

Example of Branching and Merging

```
git-graph> ./git-graph -s bold -n 30
```



The graph shows a vertical line of commits. A merge arrow points from a commit (87b4473) to the HEAD. Another merge arrow points from a commit (12a6a2b) to the HEAD. A merge arrow points from a commit (4ab7bf0) to the HEAD. A merge arrow points from a commit (49ed50a) to the HEAD. A merge arrow points from a commit (ffa542a) to the HEAD. A merge arrow points from a commit (c2181be) to the HEAD. A merge arrow points from a commit (c41eb55) to the HEAD.

```
87b4473 (HEAD -> master, origin/master) Merge pull request #135  
from peso/refac/fix-too-many-lines  
12a6a2b Merge branches 'refac/format_commit',  
'refac/print_unicode' and 'refac/from_args' into HEAD  
79ac33b Configure clippy to warn on complex functions  
f29237c Split fn format_commit  
4ab7bf0 Merge pull request #131 from mlang-42/feature/no-pager  
8aca907 Remove pager (close #107, #121)  
c712b97 Mention Homebrew installer  
49ed50a [v0.7.0] Increment version to 0.7.0  
ffa542a Fix #83: Large memory usage  
c2181be Fix #76: Exit at the end  
c41eb55 bump yansi crate to 1.x, refactor for new API
```

Core Concept: Rebasing

- ▶ **Cleaning Up History:** Instead of a messy “merge commit,” Rebasing moves your entire branch to begin on the tip of the main branch.
- ▶ **The “Linear” Advantage:** It makes the project history look like a straight line, making it much easier to read and debug.
- ▶ **Pro Tip:** Use Rebase to keep your feature branch up-to-date with the latest changes from your team.

Example of Rebased Branch

```
demo_repo> git-graph
  ● 288b3c9 (HEAD -> bob/final-verses-rebased) third verse of Amazing Grace
  ● dbfa38b second verse of Amazing Grace
└─● f016c6e (master, tom/verse-formatting) Added intra-verse linebreaks
   ● fb1e35e Added compile script for poem
   ● c6f169f (bob/final-verses) third verse of Amazing Grace
   ● 11b5b9a second verse of Amazing Grace
└─● d18edce Added first verse of Amazing Grace
   ● 49214b4 added README.md
```

GitHub: The Social Layer

- ▶ **Git is the Engine; GitHub is the Dashboard.**
- ▶ **Collaboration:** Pull Requests (PRs) allow for code reviews and discussions before code is merged.
- ▶ **Remote Backup:** Your code lives in the cloud, accessible from anywhere.
- ▶ **Community:** Rife with open source software.

GitHub Actions

- ▶ Automate development processes by executing code on GitHub events.
- ▶ Example events:
 - ▶ push to branch
 - ▶ opened pull-request to branch
 - ▶ opened issue
- ▶ Useful for Continuous Integration and Deployment (CI/CD)

Example GitHub Action

```
#.github/workflows/compile-slides.yml
name: Compile Slides
on:
  push:
    branches:
      - master
permissions:
  contents: write
jobs:
  Compile-Slides:
    runs-on: ubuntu-latest
    steps:
      - name: Updating apt
        run: sudo apt-get update
      - name: Installing pandoc
        run: sudo apt-get install pandoc -y
      - name: Installing pdflatex
        run: sudo apt install texlive-latex-extra -y
      - uses: actions/checkout@v5
      - name: Compiling Slides
        run: |
          bash compile.bash
      - name: Commit slides.pdf
        run: |
          git config --local user.email "github-actions[bot]@users.noreply.github.com"
          git config --local user.name "github-actions[bot]"
          git add -f slides.pdf
          git commit -m "[chore] Compiled and saved slides" || echo "no changes"
      - name: Push Commit
        run: git push
```